

How to manage Dataproc clusters

So far, you've learned that Dataproc is a fully managed and highly scalable service that maximizes open-source data tools for batch processing, querying, streaming, and machine learning, like Apache Hadoop and Apache Spark. Dataproc is used for data lake modernization, ETL, and secure data science. You've also learned how to create a Dataproc cluster on Google Cloud's Compute Engine service. In this reading, you'll learn more about managing Dataproc clusters, including how to start and stop a cluster, and how to add a label with a cluster.

Manage clusters

Clusters are made up of nodes, or servers dedicated to hosting only the computing resources needed to run your individual cluster. When working with clusters, you may need to increase or decrease the number of nodes. When you need a job to run faster, increasing nodes increases compute power and the amount of data that can be processed in parallel. This is one aspect of cluster management. Other aspects of cluster management include starting and stopping clusters, shutting down a cluster, and adding a label to a cluster.

Start and stop a cluster

If you're not actively working with datasets in a cluster, you can stop the cluster to avoid charges for the compute resources used to run it. To stop a cluster:

1. Navigate to the Dataproc clusters page in the Google Cloud console.
2. Click STOP to stop the cluster.

Note: If you have active jobs or operations, like updates, your cluster will not stop.

To start the cluster again, you will navigate to the same page and click START.

Delete a cluster

When a cluster is no longer needed, you can permanently delete it. To delete a cluster:

1. Navigate to the Dataproc clusters page in the Google Cloud console.
2. Select the cluster by clicking the box next to the name.
3. Click DELETE in the top menu bar.

Clusters						
+ CREATE CLUSTER REFRESH START STOP DELETE						
Filter Search clusters, press Enter						
☒	Name ↑	Status	Region	Zone	Total worker nodes	Flexible VMs?
☒	mycluster	Running	us-west1	us-west1-c	0	No

Add a label to a cluster

When you're working with multiple clusters, cluster labels can help you to organize your workloads. Labels are key-value pairs which can be assigned to Dataproc clusters and jobs. Labels enable you to sort and filter clusters. You can label clusters by project, user, or any other categories your team needs, or label if they're active or inactive.

You can use the Google Cloud console to add labels to a Dataproc cluster at any time before or after creation, and you can add a job label only at submission time. You also have the option of enabling automatically applied labels. Once your labels have been created, you can modify them. To add or delete a label on an existing cluster, follow these steps:

1. In the Google Cloud console, navigate to a cluster and click SHOW INFO PANEL.

Clusters						
+ CREATE CLUSTER REFRESH START SHOW INFO PANEL						
Filter Search clusters, press Enter						
☒	Name ↑	Status	Region	Zone	Total worker nodes	Flexible VMs?
☒	mycluster	Running	us-west1	us-west1-c	0	No

2. Add or delete labels in the LABELS panel.

The screenshot shows the Google Cloud Clusters console. At the top, there are buttons for '+ CREATE CLUSTER', 'REFRESH', 'START', and 'HIDE INFO PANEL'. Below this is a 'Filter' section with a search bar. A table lists clusters, with 'mycluster' selected. The right-hand pane shows the 'LABELS' tab for 'mycluster', which includes a description of labels, a 'Label 1' section with a key 'goog-dataproc-autozone' and a value 'enabled'.

3. Click Save.
4. To search clusters using labels, navigate to the List clusters page and use the search bar to set your filter criteria.

The screenshot shows the Google Cloud Clusters console with the 'Filter' search bar highlighted. Below the search bar is a table of clusters. The table has columns for Name, Status, Region, Zone, Total worker nodes, and Flexible VMs?.

Name	Status	Region	Zone	Total worker nodes	Flexible VMs?
mycluster	Running	us-west1	us-west1-c	0	No

Key takeaways

Dataproc clusters make batch processing, querying, and streaming data with Dataproc efficient and organized. Along with the ability to create clusters, Dataproc also allows you to start and stop clusters, or permanently delete clusters as needed. You can also add labels to your clusters to help organize your workloads. Dataproc management happens in the Google Cloud console, making all management options possible in the console.

Resources for more information

For more information about managing Dataproc clusters, including management through the command line interface, visit the following links:

- Google Cloud provides documentation for creating and managing labels in Dataproc: <https://cloud.google.com/dataproc/docs/guides/creating-managing-labels>
- Google Cloud documentation also provides information about starting and stopping clusters in Dataproc: <https://cloud.google.com/dataproc/docs/guides/dataproc-start-stop>